

Infosys Springboard

Virtual Internship 6.0

Milestone 3 – Development Phase (Part 2 + Testing)

Infosys ElectViz Election Data Visualization on for

Media

Project Title: 2024 Indian General Election

Platform: Power BI

Dataset Source: Kaggle – Lok Sabha Election 2024 Dataset
(<https://www.kaggle.com/datasets/rubenmukherjee/india-2024-election-dataset-all-candidates>)

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1.Introduction

Milestone 3 focused on completing all remaining dashboard pages, adding interactive features, finalizing DAX measures, and beginning testing to ensure accuracy and usability.

Three major analytical pages were developed:

1. Overview Analysis Page
2. State Demographics Page
3. Constituency Analysis Page

These dashboards collectively provide a detailed, multi-level analysis of the 2024 Lok Sabha Election dataset.

2.Objectives

The primary objective of Milestone 3 is to complete the development of the analytical dashboard by enhancing its functionality, accuracy, and user experience. This milestone focuses on transforming the partially built pages into fully interactive and insight-driven reports. The key objectives include:

- Develop and finalize Pages 1–3 with focused analysis views.
- Complete advanced DAX measures required for insightful analytics.
- Add interactive features to improve user exploration and storytelling.
- Perform systematic testing (data accuracy, UI, performance).
- Prepare final review and packaging for presentation/deployment.

3.Work Completed in Milestone 3

3.1 Dashboard Page 1 – Overview Analysis

Objective:
To give a national-level summary of election performance by alliances and political parties.

Key Features Developed

- KPI Cards displaying:
 - Total Seats (543)
 - NDA Seats – 292 (54%)
 - INDIA Seats – 234 (43%)
 - Other Seats – 17 (3%)
- Semi-circular arc visual showing total Parliament seats.
- Alliance-wise summary with logos and percentages.
- Party-wise breakdown for NDA, INDIA, and OTHER groups.
- Interactive tooltips for party-wise performance.
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Value Added

- Enables users to understand the national results at a glance.
- Easy comparison of major alliances.
- Strong visual representation with clean color schemes.

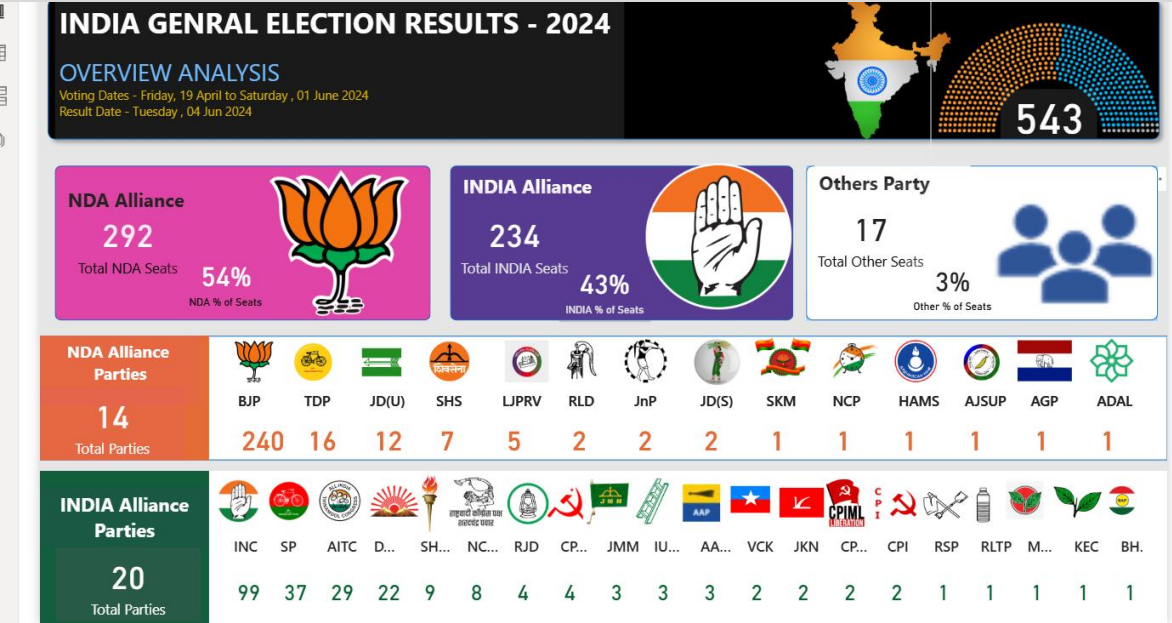


Fig no 1: Overview Analysis

3.2 Dashboard Page 2 – State Demographics

Objective:

To provide state-level and geographical insights using map-based analytics.

Key Visuals Implemented

1. State-wise Seat Distribution Map

- Shows total seats, NDA seats, and INDIA seats for each state.
- Hover interaction reveals detailed seat data
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2. Constituency and Winning Candidate Map

- Displays winning party, candidate, total votes, and margin for each constituency.
- Supports cross-filtering when a state is selected.

3. State-wise Winning Alliance Map

- Saffron color → NDA-majority states
- Blue color → INDIA-majority states
- Helps identify regional strongholds.

Value Added

- Provides spatial understanding of voting patterns.
- Shows political dominance at a state level.
- Offers drill-down access to constituency insights via map interaction.

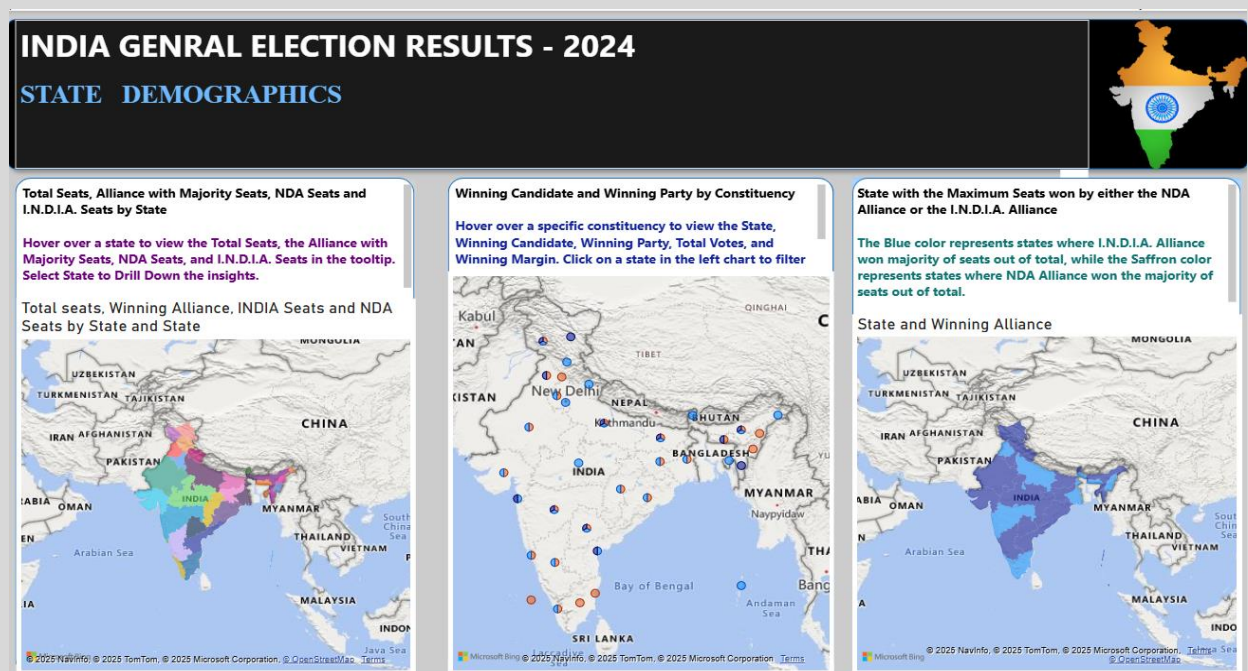


Fig no 2: State Demographics

3.3 Dashboard Page 3 – Constituency Analysis

Objective:

To create a micro-level analytical page for detailed constituency-based insights.

Key Features Developed

- Dropdown filter for selecting any constituency.
- **KPI cards showing:**
 - Total Votes
 - EVM Votes
 - Postal Votes
 - Total Candidates Participated
- **Detailed candidate cards:**
 - Winner: Name, Party, Vote Share, Total Votes
 - Runner-up: Name, Party, Vote Share
 - 2nd Runner-up: Name, Party, Vote Share
- Color-coded cards for better readability.

Value Added

- Provides a detailed breakdown of election results for each constituency.
- Helps understand voting patterns, competition levels, and margin of victory.
- Supports decision-making for political analysts and researchers.

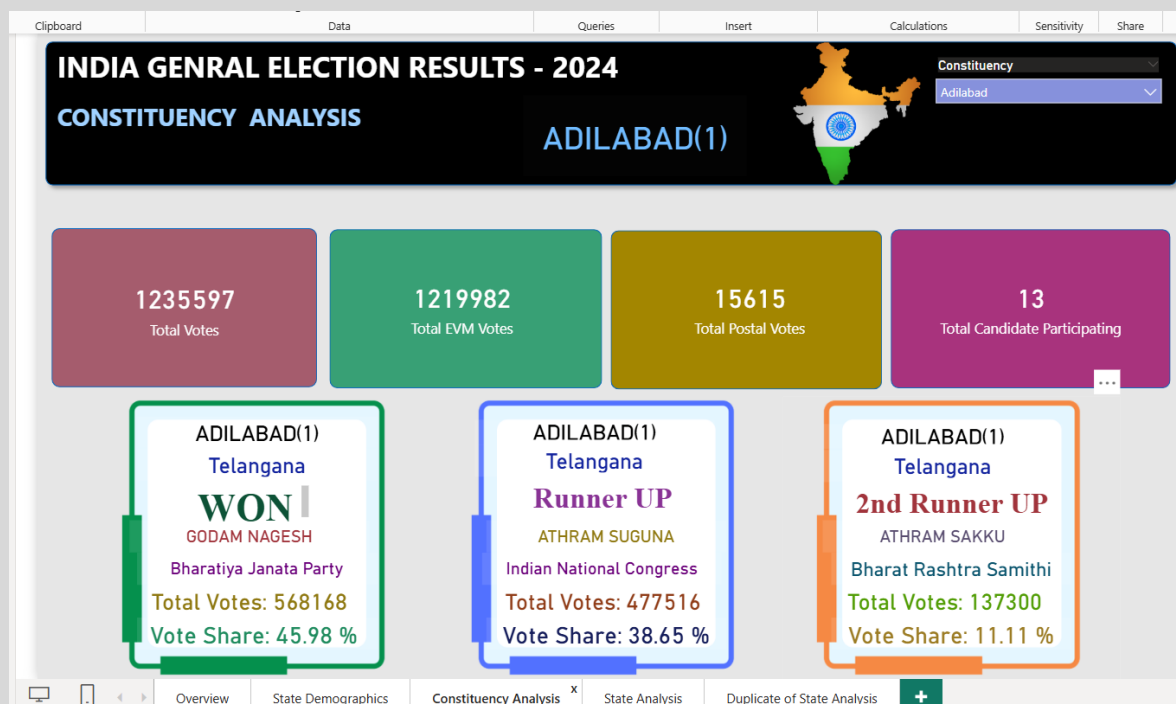


Fig no 3: Constituency Analysis

4.Interactivity Added

During Milestone 3, major interactivity features were implemented:

✓ Cross-Filtering Between Visuals

- Clicking on a state in the map instantly updates the other visuals (party seats, alliance comparison, demographics).
- Constituency markers on maps dynamically filter candidate details and vote counts.
- Alliance cards, party logos, and chart elements also act as filters for deeper exploration.

✓ Slicers for Dynamic Filtering

- State Slicer allows users to filter visuals to a specific state, enabling state-level analysis of seats and party performance.
- Constituency Slicer enables detailed drill-down to individual constituency results, including winners, runners-up, total votes, and vote share.
- Filters automatically update all visuals on the page, maintaining data consistency.

✓ Drill-Down & Hierarchical Navigation

- Users can move from National → State → Constituency levels through drill-down functionality.
- Enables multi-layered exploration while keeping the report structure clean.

✓ Tooltip-Based Insights

- Hovering over states or constituencies displays detailed information (such as total seats, winning party, winning margin, total votes).
- Tooltips help deliver insights without cluttering the dashboard visuals.

✓ Live page navigation between all 3 dashboards

This significantly improves user experience and analytical depth.

❖ Result

These interactive features significantly improve user engagement and make the dashboard more analytical, flexible, and decision-supportive. Users can derive insights instantly by interacting with any visual element.

5.DAX Measures Completed

All required DAX calculations were completed in this milestone, including:

- Total Seats
- Alliance-wise Seat Count
- Vote Share (%)
- Total Votes
- Margin Calculation
- Filtering Measures for Map Visuals

These measures support accurate, dynamic reporting across all dashboards.

6.Testing Performed

During Milestone 3, detailed testing was conducted to ensure that the dashboards were accurate, interactive, and user-ready. The testing focused on validating both the data accuracy and the functionality of all interactive components across the three pages—Overview, State Demographics, and Constituency Analysis.

Testing was conducted to ensure dashboard accuracy and performance:

✓ Functional Testing

- All visuals respond correctly to slicers.
- Maps filter properly by state and constituency.
- KPIs display accurate totals.

✓ Data Validation Testing

- Cross-checked seat counts and vote totals against raw dataset.
- Cross-checked all numerical values such as total seats, vote counts, and vote shares.
- Verified correctness of DAX measures used to calculate totals, percentages, and rankings.
- Ensured no mismatch between table data and visual outputs.

✓ Usability Testing

- Verified navigation flow between pages.
- Checked readability of cards and maps.

✓ Performance Testing

- Verified that dashboards load quickly without lag.
- Ensured optimized DAX calculations to avoid performance issues

Conclusion

Milestone 3 successfully delivered the final interactive dashboards of the Election Analysis project. All three pages—Overview, State Demographics, and Constituency Analysis—are fully functional, visually polished, and supported with accurate DAX calculations. During this milestone, significant progress was achieved in enhancing interactivity, refining the user experience, and ensuring data accuracy through comprehensive DAX calculations. The incorporation of slicers, drill-down features, cross-filtering, and dynamic maps has made the dashboard highly intuitive and user-friendly. Each visual element is backed by validated data and robust DAX measures, ensuring reliable and meaningful insights for end users. This milestone marks the near completion of the analytical development phase, with dashboards ready for final integration and review.